Uppgift 1

Original question

Låt A och B vara två händelser sådana att $P(A^c \mid B^c) = 0.6$, $P(B^c \mid A^c) = 0.4$ och $P(A^c \cup B^c) = 0.8$. Beräkna $P(A)$ och $P(B)$. (6p)

Compiled notes from Yacoub + Obid

What we are doing

We are given conditional probabilities with complements: A^c and B^c.
The notes first rewrite the conditional probabilities using the formula

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}.$$

Then we use that

$$A^c \cap B^c = B^c \cap A^c$$

is the same intersection.
Finally we use the union formula:

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c).$$

Given information

From the question:

$$P(A^c \mid B^c) = 0.6$$

$$P(B^c \mid A^c) = 0.4$$

$$P(A^c \cup B^c) = 0.8$$

We want:

$$P(A) \quad \text{and} \quad P(B).$$

Step-by-step solution

1. Rewrite the first conditional probability
Using the conditional probability formula:

$$P(A^c \mid B^c) = \frac{P(A^c \cap B^c)}{P(B^c)}$$
$$= 0.6$$

Therefore:

$$P(A^c \cap B^c) = 0.6P(B^c)$$

2. Rewrite the second conditional probability

$$P(B^c \mid A^c) = \frac{P(B^c \cap A^c)}{P(A^c)}$$
$$= 0.4$$

Therefore:

$$P(B^c \cap A^c) = 0.4P(A^c)$$

But this is the same intersection:

$$P(A^c \cap B^c) = P(B^c \cap A^c).$$

So:

$$0.6P(B^c) = 0.4P(A^c)$$

3. Isolate one complement probability

The notes isolate it in two useful ways.

First:

$$\begin{aligned} P(B^c) &= \frac{0.4}{0.6}P(A^c) \\ &= \frac{2}{3}P(A^c) \end{aligned}$$

Also:

$$\begin{aligned} P(A^c) &= \frac{0.6}{0.4}P(B^c) \\ &= \frac{3}{2}P(B^c) \end{aligned}$$

4. Use the union formula to find $P(A)$

The union formula is:

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c)$$

Use

$$P(B^c) = \frac{2}{3}P(A^c)$$

and

$$P(A^c \cap B^c) = 0.4P(A^c).$$

Then:

$$\begin{aligned} 0.8 &= P(A^c) + \frac{2}{3}P(A^c) - 0.4P(A^c) \\ 0.8 &= P(A^c)\left(1 + \frac{2}{3} - 0.4\right) \end{aligned}$$

Since $0.4 = \frac{2}{5}$:

$$\begin{aligned} 1 + \frac{2}{3} - \frac{2}{5} &= \frac{15}{15} + \frac{10}{15} - \frac{6}{15} \\ &= \frac{19}{15} \end{aligned}$$

So:

$$\begin{aligned} 0.8 &= P(A^c)\frac{19}{15} \\ P(A^c) &= \frac{0.8}{19/15} \\ &= \frac{12}{19} \end{aligned}$$

Therefore:

$$
\begin{aligned}
P(A) &= 1 - P(A^c) \\
&= 1 - \frac{12}{19} \\
&= \frac{7}{19} \\
&\approx 0.37
\end{aligned}
$$

5. Use the union formula to find $P(B)$

Now use the other replacement:

$$P(A^c) = \frac{3}{2}P(B^c)$$

and

$$P(A^c \cap B^c) = 0.6P(B^c).$$

Then:

$$
\begin{aligned}
0.8 &= \frac{3}{2}P(B^c) + P(B^c) - 0.6P(B^c) \\
0.8 &= P(B^c)\left(\frac{3}{2} + 1 - 0.6\right)
\end{aligned}
$$

The factor is:

$$
\begin{aligned}
\frac{3}{2} + 1 - 0.6 &= 1.5 + 1 - 0.6 \\
&= 1.9
\end{aligned}
$$

So:

$$
\begin{aligned}
0.8 &= 1.9P(B^c) \\
P(B^c) &= \frac{0.8}{1.9} \\
&= \frac{8}{19}
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
P(B) &= 1 - P(B^c) \\
&= 1 - \frac{8}{19} \\
&= \frac{11}{19} \\
&\approx 0.58
\end{aligned}
$$

Final answer

$$P(A) = \frac{7}{19} \approx 0.37.$$

$$P(B) = \frac{11}{19} \approx 0.58.$$

What to remember

Conditional probability:

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}.$$

Union formula:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

Here the key trick is that

$$A^c \cap B^c = B^c \cap A^c$$

is the same overlap, so the two conditional formulas can be connected.

Uppgift 2

Original question

Betrakta funktionen

$$f(x) = \begin{cases} ax^2, & 0 < x < 3, \\ 0, & \text{annars.} \end{cases}$$

a) Bestäm a så att f bildar en täthetsfunktion för en s.v. X. (1.5p)

b) Bestäm fördelningsfunktionen F_X. (1.5p)

c) Beräkna $p(X > 1)$. (1p)

d) Beräkna väntevärde och standardavvikelse för X. (2p)

Compiled notes from Yacoub + Obid

What we are doing

The notes say that f is a density function, or PDF. A PDF represents the density, and the total area under the graph must be 1.

First we choose a so that the total area is 1. Then we integrate the PDF from the lower limit up to x to get the CDF. After that we use the CDF to calculate $P(X > 1)$. Finally we calculate the expected value and standard deviation.

Given information

The function is:

$$f(x) = \begin{cases} ax^2, & 0 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

The interval where the density is nonzero is:

$$0 < x < 3.$$

Step-by-step solution

a) Find a

For a PDF, the total area must be 1:

$$\int_0^3 ax^2 \, dx = 1.$$

Integrate:

$$\int_0^3 ax^2 \, dx = \left[a\frac{x^3}{3} \right]_0^3$$

$$= 1$$

Plug in the limits:

$$a\frac{3^3}{3} - 0 = 1$$

$$a\frac{27}{3} = 1$$

$$9a = 1$$

$$a = \frac{1}{9}$$

So the PDF is:

$$f(x) = \frac{1}{9}x^2, \qquad 0 < x < 3.$$

Original question repeated for this part

Betrakta funktionen

$$f(x) = \begin{cases} ax^2, & 0 < x < 3, \\ 0, & \text{annars.} \end{cases}$$

a) Bestäm a så att f bildar en täthetsfunktion för en s.v. X. (1.5p)

b) Bestäm fördelningsfunktionen F_X. (1.5p)

c) Beräkna $p(X > 1)$. (1p)

d) Beräkna väntevärde och standardavvikelse för X. (2p)

b) Find the CDF

The notes say the CDF is the probability up to a point x. The PDF represents the whole graph, while the CDF accumulates the area up to x.

For $0 < x < 3$:

$$\begin{aligned} F_X(x) &= \int_0^x \frac{1}{9}t^2\,dt \\ &= \frac{1}{9}\left[\frac{t^3}{3}\right]_0^x \\ &= \frac{1}{9}\left(\frac{x^3}{3} - 0\right) \\ &= \frac{x^3}{27} \end{aligned}$$

So:

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ \dfrac{x^3}{27}, & 0 < x < 3, \\ 1, & x \ge 3. \end{cases}$$

Original question repeated for this part

Betrakta funktionen

$$f(x) = \begin{cases} ax^2, & 0 < x < 3, \\ 0, & \text{annars.} \end{cases}$$

a) Bestäm a så att f bildar en täthetsfunktion för en s.v. X. (1.5p)

b) Bestäm fördelningsfunktionen F_X. (1.5p)

c) Beräkna $p(X > 1)$. (1p)

d) Beräkna väntevärde och standardavvikelse för X. (2p)

c) Calculate $P(X > 1)$

The notes explain that $P(X > 1)$ means the chance for the outcome of the random variable to be bigger than 1.

Use the CDF:

$$P(X > 1) = 1 - P(X \leq 1)$$
$$= 1 - F_X(1)$$

From part b:

$$F_X(x) = \frac{x^3}{27}.$$

Plug in $x = 1$:

$$F_X(1) = \frac{1^3}{27}$$
$$= \frac{1}{27}$$

Therefore:

$$P(X > 1) = 1 - \frac{1}{27}$$
$$= \frac{26}{27}$$
$$\approx 0.963$$

So the chance that the random variable gives an outcome bigger than 1 is about 96.3 %.

Original question repeated for this part

Betrakta funktionen

$$f(x) = \begin{cases} ax^2, & 0 < x < 3, \\ 0, & \text{annars.} \end{cases}$$

a) Bestäm a så att f bildar en täthetsfunktion för en s.v. X. (1.5p)

b) Bestäm fördelningsfunktionen F_X. (1.5p)

c) Beräkna $p(X > 1)$. (1p)

d) Beräkna väntevärde och standardavvikelse för X. (2p)

d) Expected value and standard deviation

The notes describe expected value as the center, or mean:

$$\mu = E[X].$$

Standard deviation tells how much the values vary from the mean.

Find $E[X]$

The notes say to multiply x with the PDF:

$$E[X] = \int_0^3 x \left(\frac{1}{9}x^2\right) dx$$
$$= \int_0^3 \frac{1}{9}x^3 \, dx$$
$$= \frac{1}{9}\left[\frac{x^4}{4}\right]_0^3$$
$$= \frac{1}{9}\left(\frac{81}{4} - 0\right)$$
$$= \frac{81}{36}$$
$$= \frac{9}{4}$$

$$= 2.25$$

So:

$$E[X] = 2.25.$$

Find $E[X^2]$

For standard deviation, first use:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2.$$

The notes say to multiply x^2 with the PDF:

$$\begin{aligned} E[X^2] &= \int_0^3 x^2 \left(\frac{1}{9}x^2\right) dx \\ &= \int_0^3 \frac{1}{9}x^4 \, dx \\ &= \frac{1}{9}\left[\frac{x^5}{5}\right]_0^3 \\ &= \frac{1}{9}\left(\frac{243}{5} - 0\right) \\ &= 5.4 \end{aligned}$$

Now calculate the variance:

$$\begin{aligned} \mathrm{Var}(X) &= E[X^2] - (E[X])^2 \\ &= 5.4 - (2.25)^2 \\ &= 5.4 - 5.0625 \\ &= 0.3375 \end{aligned}$$

Then:

$$\begin{aligned} \sigma &= \sqrt{\mathrm{Var}(X)} \\ &= \sqrt{0.3375} \\ &\approx 0.581 \end{aligned}$$

Final answer

a)

$$a = \frac{1}{9}.$$

b)

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ \dfrac{x^3}{27}, & 0 < x < 3, \\ 1, & x \ge 3. \end{cases}$$

c)

$$P(X > 1) = \frac{26}{27} \approx 0.963.$$

d)

$$E[X] = 2.25, \qquad \sigma \approx 0.581.$$

What to remember

For a density function:

$$\int f(x)\, dx = 1.$$

For a CDF:

$$F_X(x) = P(X \leq x).$$

For expected value:

$$E[X] = \int x f(x)\, dx.$$

For standard deviation:

$$\sigma = \sqrt{E[X^2] - (E[X])^2}.$$

Uppgift 3

Original question

Låt $X_1, X_2 \in \mathrm{P}(\mu)$, där $\mu = 2$ och antag att X_1, X_2 är oberoende s.v.

a) Beräkna $P(X_1 = 1)$ och $P(1 \leq X_1 < 4)$.

b) Beräkna $P(\max(X_1, X_2) \leq 2)$.

c) Låt $Y = X_1 - 2X_2$. Beräkna $E(Y)$ och $V(Y)$.

Compiled notes from Yacoub + Obid

What we are doing

The notes use the Poisson distribution. They describe it as a tool to predict the odds of random fluctuations.

If we know the mean, the Poisson formula gives the chance that an exact event happens k times in a specific window.

The notes also remind us that for a Poisson random variable:

$$E[X] = \mu, \qquad \mathrm{Var}(X) = \mu.$$

Here:

$$\mu = 2.$$

Given information

$$X_1, X_2 \in \mathrm{P}(2)$$

$$X_1 \text{ and } X_2 \text{ are independent.}$$

The Poisson formula is:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}.$$

With $\mu = 2$:

$$P(X = k) = \frac{2^k e^{-2}}{k!}.$$

Step-by-step solution

a) Calculate $P(X_1 = 1)$

Use the formula with $k = 1$:

$$\begin{aligned} P(X_1 = 1) &= \frac{2^1 e^{-2}}{1!} \\ &= 2e^{-2} \\ &\approx 0.2707 \end{aligned}$$

Calculate $P(1 \leq X_1 < 4)$

Since X_1 is discrete:

$$1 \leq X_1 < 4$$

means:

$$X_1 = 1, 2, 3.$$

So:

$$P(1 \leq X_1 < 4) = P(X_1 = 1) + P(X_1 = 2) + P(X_1 = 3)$$

The notes write:

$$P(X_1 = 1) \approx 0.2707.$$

For $X_1 = 2$:

$$\begin{aligned} P(X_1 = 2) &= \frac{2^2 e^{-2}}{2!} \\ &= 2e^{-2} \\ &\approx 0.2707 \end{aligned}$$

For $X_1 = 3$:

$$\begin{aligned} P(X_1 = 3) &= \frac{2^3 e^{-2}}{3!} \\ &= \frac{8}{6} e^{-2} \\ &\approx 0.1804 \end{aligned}$$

Therefore:

$$\begin{aligned} P(1 \leq X_1 < 4) &\approx 0.2707 + 0.2707 + 0.1804 \\ &= 0.7218 \end{aligned}$$

Original question repeated for this part

Låt $X_1, X_2 \in \mathrm{P}(\mu)$, där $\mu = 2$ och antag att X_1, X_2 är oberoende s.v.

a) Beräkna $P(X_1 = 1)$ och $P(1 \leq X_1 < 4)$.

b) Beräkna $P(\max(X_1, X_2) \leq 2)$.

c) Låt $Y = X_1 - 2X_2$. Beräkna $E(Y)$ och $V(Y)$.

b) Calculate $P(\max(X_1, X_2) \leq 2)$

The notes explain this as: both X_1 and X_2 must be at most 2.

So:

$$\max(X_1, X_2) \leq 2$$

means:

$$X_1 \leq 2 \quad \text{and} \quad X_2 \leq 2.$$

Because X_1 and X_2 are independent:

$$P(\max(X_1, X_2) \leq 2) = P(X_1 \leq 2) \cdot P(X_2 \leq 2)$$

Find $P(X \leq 2)$:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

The notes calculate:

$$P(X = 0) = \frac{2^0 e^{-2}}{0!} = e^{-2}$$
$$P(X = 1) = 2e^{-2}$$
$$P(X = 2) = 2e^{-2}$$

So:

$$P(X \leq 2) = e^{-2} + 2e^{-2} + 2e^{-2}$$
$$= 5e^{-2}$$

Thus:

$$P(\max(X_1, X_2) \leq 2) = 5e^{-2} \cdot 5e^{-2}$$
$$= 25e^{-4}$$
$$\approx 0.4578$$

Original question repeated for this part

Låt $X_1, X_2 \in \mathrm{P}(\mu)$, där $\mu = 2$ och antag att X_1, X_2 är oberoende s.v.

a) Beräkna $P(X_1 = 1)$ och $P(1 \leq X_1 < 4)$.

b) Beräkna $P(\max(X_1, X_2) \leq 2)$.

c) Låt $Y = X_1 - 2X_2$. Beräkna $E(Y)$ och $V(Y)$.

c) Calculate $E(Y)$ and $V(Y)$

The variable is:

$$Y = X_1 - 2X_2.$$

The notes split the plus/minus parts in the variable.

For expectation:

$$E(aX_1 + bX_2) = aE[X_1] + bE[X_2].$$

So:

$$E(Y) = E(X_1 - 2X_2)$$
$$= E(X_1) - 2E(X_2)$$
$$= 1 \cdot 2 - 2 \cdot 2$$
$$= 2 - 4$$
$$= -2$$

For variance, since X_1 and X_2 are independent:

$$\mathrm{Var}(aX_1 + bX_2) = a^2 \,\mathrm{Var}(X_1) + b^2 \,\mathrm{Var}(X_2).$$

Use $\mathrm{Var}(X_1) = 2$ and $\mathrm{Var}(X_2) = 2$:

$$V(Y) = \mathrm{Var}(X_1 - 2X_2)$$
$$= 1^2 \,\mathrm{Var}(X_1) + (-2)^2 \,\mathrm{Var}(X_2)$$
$$= 1^2 \cdot 2 + 4 \cdot 2$$
$$= 2 + 8$$
$$= 10$$

Final answer

a)

$$P(X_1 = 1) = 2e^{-2} \approx 0.2707.$$

$$P(1 \leq X_1 < 4) \approx 0.7218.$$

b)

$$P(\max(X_1, X_2) \leq 2) = 25e^{-4} \approx 0.4578.$$

c)

$$E(Y) = -2, \qquad V(Y) = 10.$$

What to remember

For Poisson:

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!}.$$

For Poisson:

$$E[X] = \mu, \qquad \mathrm{Var}(X) = \mu.$$

For independent variables:

$$P(X_1 \leq 2,\ X_2 \leq 2) = P(X_1 \leq 2)P(X_2 \leq 2).$$

For variance of a linear combination:

$$\mathrm{Var}(aX_1 + bX_2) = a^2\,\mathrm{Var}(X_1) + b^2\,\mathrm{Var}(X_2).$$

Uppgift 4

Original question

Låt $x_1, x_2, \ldots, x_n$, där $0 < x_i < 1$ för alla i, vara ett stickprov från en fördelning med täthetsfunktion

$$f_X(x) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1, \\ 0, & \text{annars.} \end{cases}$$

Ange ML-skattningen av θ. (6p)

Compiled notes from Yacoub + Obid

What we are doing

The notes say we want to find the formula for θ that maximizes the probability of getting exactly the data we observed.

This is maximum likelihood:

- build the likelihood function,
- take the logarithm,
- differentiate,
- set the derivative equal to zero,
- solve for θ.

Given information

The data are:

$$x_1, x_2, \ldots, x_n$$

and each value satisfies:

$$0 < x_i < 1.$$

The density is:

$$f_X(x) = \theta x^{\theta-1}, \qquad 0 < x < 1.$$

Step-by-step solution

Step 1: Likelihood function

The notes use:

$$L(\theta) = \prod_{i=1}^{n} f_X(x_i).$$

Substitute the density:

$$L(\theta) = \prod_{i=1}^{n} \left(\theta x_i^{\theta-1}\right)$$

Apply the product rule:

$$L(\theta) = \left(\prod_{i=1}^{n} \theta\right) \left(\prod_{i=1}^{n} x_i\right)^{\theta-1}$$

Since there are n copies of θ:

$$L(\theta) = \theta^n \left(\prod_{i=1}^{n} x_i\right)^{\theta-1}$$

Step 2: Log-likelihood

The notes take the logarithm:

$$\ln L(\theta) = \ln \left[\theta^n \left(\prod_{i=1}^{n} x_i\right)^{\theta-1}\right]$$

Use:

$$\ln(A \cdot B) = \ln(A) + \ln(B)$$

and:

$$\ln(A^b) = b \ln(A).$$

Then:

$$\begin{aligned} \ln L(\theta) &= \ln(\theta^n) + \ln \left[\left(\prod_{i=1}^{n} x_i\right)^{\theta-1}\right] \\ &= n \ln \theta + (\theta - 1) \ln \left(\prod_{i=1}^{n} x_i\right) \end{aligned}$$

The notes also use:

$$\ln \left(\prod x_i\right) = \sum \ln(x_i).$$

So:

$$\ln L(\theta) = n \ln \theta + (\theta - 1) \sum_{i=1}^{n} \ln(x_i)$$

Step 3: Differentiate

To find the maximum, take the derivative and set it equal to zero:

$$\frac{d}{d\theta} \ln L(\theta) = \frac{d}{d\theta} \left[n \ln \theta + (\theta - 1) \sum_{i=1}^{n} \ln(x_i) \right]$$

Differentiate:

$$\frac{d}{d\theta} \ln L(\theta) = \frac{n}{\theta} + 1 \cdot \sum_{i=1}^{n} \ln(x_i)$$

Step 4: Set equal to zero and solve

$$\frac{n}{\theta} + \sum_{i=1}^{n} \ln(x_i) = 0$$

Move the sum to the other side:

$$\frac{n}{\theta} = -\sum_{i=1}^{n} \ln(x_i)$$

Solve for θ:

$$\theta = \frac{n}{-\sum_{i=1}^{n} \ln(x_i)}$$

Final answer

$$\hat{\theta}_{\text{ML}} = \frac{n}{-\sum_{i=1}^{n} \ln(x_i)}.$$

What to remember

For ML estimation:

$$L(\theta) = \prod_{i=1}^{n} f_X(x_i).$$

Taking logs turns products into sums:

$$\ln \left(\prod x_i \right) = \sum \ln(x_i).$$

To maximize, differentiate the log-likelihood and set it equal to zero.

Uppgift 5

Original question

Följande 8 observationer kan antas vara observationer från $N(\mu, \sigma)$:

$$13.4,\ 14.1,\ 15.1,\ 14.3,\ 14.9,\ 13.9,\ 14.4,\ 14.0.$$

Beräkning ger att

$$\sum_{i=1}^{8} x_i = 114.1, \qquad \sum_{i=1}^{8} x_i^2 = 1629.45.$$

a) Ange en lämplig punktskattning av σ^2 och σ. (2p)

b) Ange ett 90 % konfidensintervall för σ. (4p)

Compiled notes from Yacoub + Obid

What we are doing

The notes use descriptive statistics for a normal sample. First we estimate the variance and standard deviation from the sample.

Then we build a 90 % confidence interval for σ. The notes first find the interval for σ^2 using the chi-square distribution, and then take square roots to get the interval for σ.

Given information

From the question:

$$n = 8$$

$$\sum x_i = 114.1$$

$$\sum x_i^2 = 1629.45$$

For part b:

$$90\,\% \text{ confidence}$$

so

$$\alpha = 0.10.$$

Step-by-step solution

a) Point estimate of σ^2 and σ

The notes use the sample variance:

$$s^2 = \frac{1}{n-1}\left(\sum x_i^2 - n\bar{x}^{\,2}\right).$$

Step 1: Find the sample mean

$$\begin{aligned}\bar{x} &= \frac{\sum x_i}{n} \\ &= \frac{114.1}{8} \\ &= 14.2625 \\ &\approx 14.26\end{aligned}$$

Step 2: Calculate the numerator

$$\begin{aligned}\sum x_i^2 - n\bar{x}^{\,2} &= 1629.45 - 8(14.2625)^2 \\ &= 2.09875\end{aligned}$$

Step 3: Divide by $n-1$

$$\begin{aligned}s^2 &= \frac{2.09875}{8-1} \\ &= \frac{2.09875}{7} \\ &\approx 0.2998\end{aligned}$$

So the point estimate of σ^2 is:

$$s^2 \approx 0.2998.$$

For σ, take the square root:

$$s = \sqrt{0.2998}$$

$$\approx 0.5476$$

So the point estimate of σ is:

$$s \approx 0.5476.$$

Original question repeated for this part

Följande 8 observationer kan antas vara observationer från $N(\mu, \sigma)$:

$$13.4,\ 14.1,\ 15.1,\ 14.3,\ 14.9,\ 13.9,\ 14.4,\ 14.0.$$

Beräkning ger att

$$\sum_{i=1}^{8} x_i = 114.1, \qquad \sum_{i=1}^{8} x_i^2 = 1629.45.$$

a) Ange en lämplig punktskattning av σ^2 och σ. (2p)

b) Ange ett 90 % konfidensintervall för σ. (4p)

b) 90% confidence interval for σ

The notes use the chi-square confidence interval for variance:

$$\left[\frac{(n-1)s^2}{\chi^2_{\text{upper}}}, \frac{(n-1)s^2}{\chi^2_{\text{lower}}}\right].$$

Here:

$$df = n - 1 = 8 - 1 = 7.$$

For a 90 % confidence interval:

$$\alpha = 0.10.$$

Split the error into two tails:

$$0.10/2 = 0.05.$$

From the chi-square table with $df = 7$, the notes use:

$$\chi^2_{0.95} = 14.067$$

$$\chi^2_{0.05} = 2.167$$

Also, from part a:

$$(n-1)s^2 = 2.09875.$$

Lower limit for σ^2

$$\frac{(n-1)s^2}{\chi^2_{0.95}} = \frac{2.09875}{14.067}$$

$$\approx 0.1492$$

Upper limit for σ^2

$$\frac{(n-1)s^2}{\chi^2_{0.05}} = \frac{2.09875}{2.167}$$

$$\approx 0.9685$$

So the confidence interval for the variance is:

$$\sigma^2 \in [0.1492,\ 0.9685].$$

To get the interval for σ, take square roots:

$$\sigma \in \left[\sqrt{0.1492}, \sqrt{0.9685}\right]$$

$$\approx [0.386,\ 0.984].$$

Final answer

a)

$$\widehat{\sigma^2} = s^2 \approx 0.2998.$$

$$\hat{\sigma} = s \approx 0.5476.$$

b)

90 % confidence interval for σ :

$$\sigma \in [0.386,\ 0.984].$$

What to remember

Sample variance:

$$s^2 = \frac{1}{n-1}\left(\sum x_i^2 - n\bar{x}^{\,2}\right).$$

For a normal sample, a confidence interval for σ^2 uses the chi-square distribution:

$$\left[\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{\alpha/2}}\right].$$

For σ, take the square root of both limits.

Uppgift 6

Original question

Vikten av en flygpassagerare (inklusive handbagage) antas vara normalfördelad $N(\mu, \sigma)$. Under många år antogs $\mu = 80$ kg, men man vill nu testa om medelvikten har ökat. Vi kan anta att $\sigma = 10$.

a) Undersök om $\mu > 80$ genom att ställa upp ett lämpligt hypotestest, med signifikansnivån 5 % (nollhypotes och mothypotes ska vara tydligt formulerade). (2p)

b) Utför ditt hypotestest i a) då $n = 100$ och $\bar{x} = 83$. (2p)

c) Antag att man tror att medelvikten ökat med 2 kg (alltså $\mu = 82$). Hur stort måste n väljas för att kunna upptäcka denna ökning med 80 % sannolikhet? Alltså, hur stort måste n väljas så att $P(\text{Förkasta } H_0) > 0.8$, under antagandet att $\mu = 82$? (Du kan här använda att $\Phi(0.84) \approx 0.8$.) (2p)

Compiled notes from Yacoub + Obid

What we are doing

The notes set up a hypothesis test for the mean weight. The old mean is 80 kg, and we test if the mean has increased.

Since $\sigma = 10$ is known, the notes use a normal z test.

Part c is about statistical power: we choose n large enough so that if the true mean is 82, we reject H_0 with probability at least 80 %.

Given information

$$X \sim N(\mu, \sigma)$$

$$\sigma = 10$$

Old assumed mean:

$$\mu = 80$$

Significance level:

$$\alpha = 5\,\% = 0.05$$

Step-by-step solution

a) Set up the hypothesis test

The notes write:

$$H_0 : \mu = 80$$

$$H_1 : \mu > 80$$

This is a one-sided test, because the question asks if the mean weight has increased.

The significance level is:

$$\alpha = 0.05.$$

The notes explain this as: H_0 is rejected only when there is less than a 5 % chance that the result happened by luck.

Original question repeated for this part

Vikten av en flygpassagerare (inklusive handbagage) antas vara normalfördelad $N(\mu, \sigma)$. Under många år antogs $\mu = 80$ kg, men man vill nu testa om medelvikten har ökat. Vi kan anta att $\sigma = 10$.

a) Undersök om $\mu > 80$ genom att ställa upp ett lämpligt hypotestest, med signifikansnivån 5 % (nollhypotes och mothypotes ska vara tydligt formulerade). (2p)

b) Utför ditt hypotestest i a) då $n = 100$ och $\bar{x} = 83$. (2p)

c) Antag att man tror att medelvikten ökat med 2 kg (alltså $\mu = 82$). Hur stort måste n väljas för att kunna upptäcka denna ökning med 80 % sannolikhet? Alltså, hur stort måste n väljas så att $P(\text{Förkasta } H_0) > 0.8$, under antagandet att $\mu = 82$? (Du kan här använda att $\Phi(0.84) \approx 0.8$.) (2p)

b) Perform the test for $n = 100$ and $\bar{x} = 83$

Given:

$$n = 100, \qquad \bar{x} = 83.$$

The standard error is:

$$\begin{aligned} SE &= \frac{\sigma}{\sqrt{n}} \\ &= \frac{10}{\sqrt{100}} \\ &= \frac{10}{10} \\ &= 1 \end{aligned}$$

The z-score is:

$$z = \frac{\bar{x} - \mu}{SE}$$

$$= \frac{83 - 80}{1}$$

$$= 3$$

For a one-sided 5 % test, the notes use the normal table:

$$z_{0.95} \approx 1.645.$$

Compare:

$$3 > 1.645.$$

So we reject H_0.

There is evidence that the average weight has increased.

Original question repeated for this part

Vikten av en flygpassagerare (inklusive handbagage) antas vara normalfördelad $N(\mu, \sigma)$. Under många år antogs $\mu = 80$ kg, men man vill nu testa om medelvikten har ökat. Vi kan anta att $\sigma = 10$.

a) Undersök om $\mu > 80$ genom att ställa upp ett lämpligt hypotestest, med signifikansnivån 5 % (nollhypotes och mothypotes ska vara tydligt formulerade). (2p)

b) Utför ditt hypotestest i a) då $n = 100$ och $\bar{x} = 83$. (2p)

c) Antag att man tror att medelvikten ökat med 2 kg (alltså $\mu = 82$). Hur stort måste n väljas för att kunna upptäcka denna ökning med 80 % sannolikhet? Alltså, hur stort måste n väljas så att $P(\text{Förkasta } H_0) > 0.8$, under antagandet att $\mu = 82$? (Du kan här använda att $\Phi(0.84) \approx 0.8$.) (2p)

c) Required sample size

The notes say this is about statistical power. We want n large enough so that, if the true mean is 82, we catch it 80 % of the time.

Given:

$$\mu = 82$$

$$P(\text{reject } H_0) > 0.8$$

and

$$\Phi(0.84) \approx 0.8.$$

The rejection rule from the one-sided test is:

$$\bar{x} > 80 + 1.645 \cdot \frac{10}{\sqrt{n}}.$$

The notes write the power requirement as:

$$P\left(\bar{x} > 80 + 1.645 \cdot \frac{10}{\sqrt{n}}\right) = 0.8, \qquad \text{given } \mu = 82.$$

Standardize using $\mu = 82$:

$$P\left(Z > \frac{\left(80 + 1.645 \cdot \frac{10}{\sqrt{n}}\right) - 82}{10/\sqrt{n}}\right) = 0.8.$$

This becomes:

$$P\left(Z > \frac{-2 + 1.645 \cdot \frac{10}{\sqrt{n}}}{10/\sqrt{n}}\right) = 0.8.$$

English clarification: The notes stop here, but the next algebra step is directly implied. Since $P(Z > -0.84) \approx 0.8$, set the inside value equal to -0.84.

$$\frac{-2 + 1.645 \cdot \frac{10}{\sqrt{n}}}{10/\sqrt{n}} = -0.84$$

Split the fraction:

$$\frac{-2}{10/\sqrt{n}} + 1.645 = -0.84$$

$$-\frac{2\sqrt{n}}{10} + 1.645 = -0.84$$

Move terms:

$$-\frac{2\sqrt{n}}{10} = -0.84 - 1.645$$

$$-\frac{2\sqrt{n}}{10} = -2.485$$

So:

$$\frac{2\sqrt{n}}{10} = 2.485$$

$$\sqrt{n} = \frac{2.485 \cdot 10}{2}$$

$$= 12.425$$

Square both sides:

$$n = (12.425)^2$$

$$\approx 154.38$$

Since n must be a whole number:

$$n = 155.$$

Final answer

a)

$$H_0 : \mu = 80, \qquad H_1 : \mu > 80.$$

b)

$$z = 3.$$

Since $3 > 1.645$, reject H_0. There is evidence that the mean weight has increased.

c)

$$n \geq 155.$$

So choose

$$n = 155.$$

What to remember

For a one-sided test with known σ:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

At significance level 5 % for an upper-tail test:

$$z_{0.95} \approx 1.645.$$

For power calculations, write the rejection rule in terms of $\bar{x}$, then standardize using the alternative mean.